Mon 24 Feb 2025

- midterm on Friday
- A-L MC 252
  M-Z 254
- see Quercus for sample tests. (they are a bit too long)
- Aids:
  - simple calculator
    - show your steps
  - aid sheet
    - handwritten
    - single sided 8½" x 11"
    - no copies

  I have a few spares.
- up to end of Wed

  (not meaning to directly test on info. mentioned today or Wed, though some things covered may help your understanding)
- textbook – use to clarify understanding of material covered in lecture.
  - has good questions at the end of the chapters.
- A2 questions?

$\boxed{\text{Proof that } \|A\|_\infty = \text{max abs row sum}}$

Let A be a $n \times n$ matrix and $\underline{x}$ a $n \times 1$ vector.

We have by def$^{\underline{n}}$ that $\|A\|_\infty = \max\limits_{\|\underline{x}\|_\infty = 1} \|A\underline{x}\|_\infty$.

and $\|\underline{x}\|_\infty = \max\limits_{1 \le i \le n} |x_i|$.

Let $\underline{x}$ be such that $\|\underline{x}\|_\infty = 1$ and write $A = [\underline{a}_1\ \underline{a}_2 \cdots \underline{a}_n]$ (column wise)

Then $A\underline{x} = x_1\underline{a}_1 + x_2\underline{a}_2 + \dots + x_n\underline{a}_n = \sum\limits_{j=1}^{n} x_j \underline{a}_j$.

And $(A\underline{x})_i = \sum\limits_{j=1}^{n} x_j a_{i,j}$. (the ith component of $A\underline{x}$)

Also, $\|A\|_\infty = \max\limits_{1 \le i \le n} |(A\underline{x})_i|$

$= \max\limits_{1 \le i \le n} \left| \sum\limits_{j=1}^{n} x_j a_{i,j} \right|$

For $\|\underline{x}\|_\infty = 1$, for any $1 \le i \le n$ $\left| \sum\limits_{j=1}^{n} x_j a_{i,j} \right|$ will be maximized by taking each $x_j = 1 \cdot \text{sign}(a_{i,j})$ (making each term in sum $+$)

Then each $x_j a_{i,j} = |a_{i,j}|$ and $\left| \sum\limits_{j=1}^{n} x_j a_{ij} \right| = \sum\limits_{j=1}^{n} |a_{i,j}|$

Hence, $\|A\|_\infty = \max\limits_{1 \le i \le n} \left(\sum\limits_{j=1}^{n} |a_{i,j}| \right)$

= the maximum absolute row sum.

Note: proof that $\|A\|_1 = \text{max abs. column sum}$ follows a similar approach

We have seen that the accuracy of our computed solution can depend on the condition number of our A matrix

When will the condition number be large?

Solving $A\underline{x} = \underline{b}$ can be thought of as finding the intersection point of a set of lines.

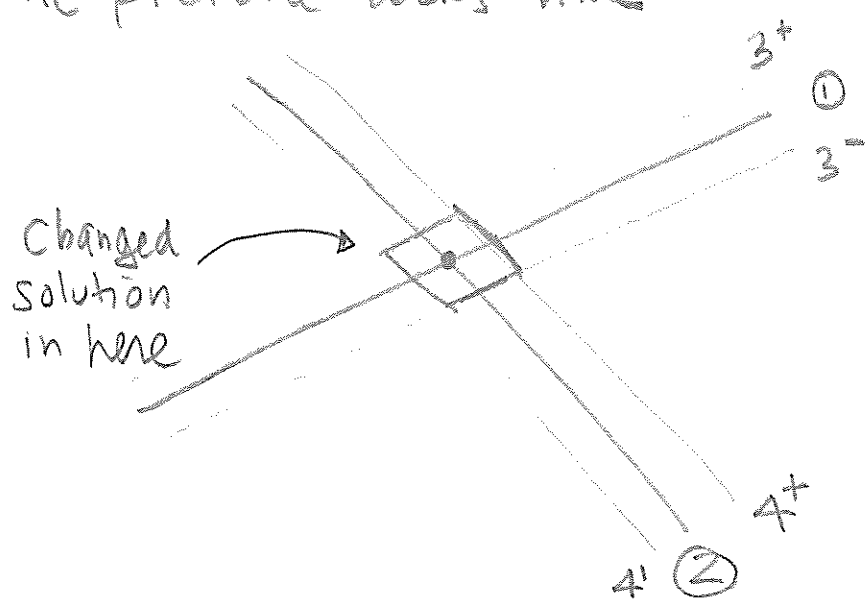
Consider the $n=2$ case and these systems

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 & (1) \\ a_{21}x_1 + a_{22}x_2 = b_2 & (2) \end{cases} \qquad \begin{cases} a_{11}x_1 + a_{12}x_2 = \hat{b}_1 & (3) \\ a_{21}x_1 + a_{22}x_2 = \hat{b}_2 & (4) \end{cases}$$

lines (1), (3) have same slope, different intercept.

lines (2), (4) have same slope, different intercept.

If the picture looks like

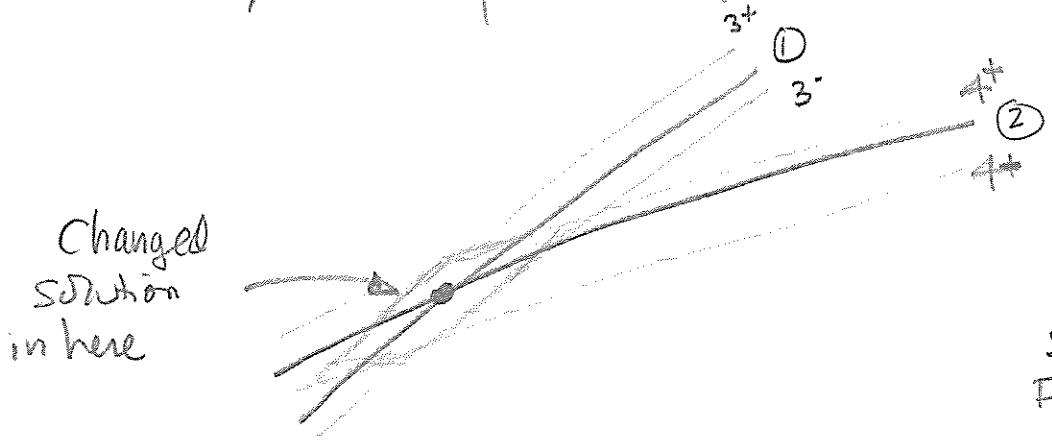


see Heath Fig 2.4

then a small change from $\underline{b}$ to $\hat{\underline{b}}$ produces a small change in the solution

Well conditioned

However, if the picture looks like



see Heath Fig 2.4

A small change in the $\underline{b}$'s can produce a large change in the solution.

ill-conditioned

The condition number of A will be large when A is close to having linearly dependent rows/columns, i.e., A is close to being singular.

To assess the accuracy of a solution, we will want to know cond(A).

We know $\text{cond}(A) = \|A\| \, \|A^{-1}\|$.

Computing $\|A\|_1$ or $\|A\|_\infty$ is easy.

What about $\|A^{-1}\|$?

Exact computation would take alot of time. Since use of cond(A) is as a diagnostic tool, we don't need to know it exactly. An estimate of the size is sufficient.